Best bets after the frontier check

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The ambitious question stays at the centre: can training make the model's self govern what it does in situations it has never seen? Two main bets test it: training a self-commitment (Main bet A) and keeping the assistant present under pressure (Main bet B). The frontier check changed *how* we run them (power, controls, models), not *what* they ask.

How we vetted

A bet was dropped only if published work already answers the same question in the same setting. A nearby precedent sharpens a bet; it does not kill it.

Each bet was sorted into one of three outcomes:

Outcome	Rule	What we do
Keep	No study answers the question directly	Run it, and cite the closest work to show the gap
Sharpen	Nearby work exists, but in a different setting (chat instead of agents, fine-tuning instead of context) or with a key control missing	Run it, framed as an extension, with the design fixes the precedent suggests
Present as a replication	The same result is already published	Don't spend new compute on it; mention it as consistent with prior work

Nothing in the plan fell into a fourth category, "already done exactly, drop it." The closest case is self-recognition training against emergent misalignment, which we never planned to rerun.

Ranking is separate from vetting. The check informs design, not priority. Nothing it found answers the training questions, so the two training bets stay the main bets, as in the original plan. The in-context tests run alongside them as diagnostics, and the salience checks are support for the slides.

The best bets at a glance

The two main bets need training. The three A0 tests run in context, alongside Main bet A, and feed it. The two support checks firm up the salience result for the slides.

Part	Bet	The question in plain words	Status	Closest precedent
Main A	**Training a self-commitment**	Does training on "forecast, then act" carry over to new tasks better than training on rules, on situation cues, or on the actions alone?	Sharpen	Training on its own record changes behaviour; reasons beat rules; self-consistency training
A0.1 (in context)	Breadth test	Does "I'm an assistant who doesn't cut corners" stop violations it never names, better than the same idea said impersonally?	Keep	Rules against values compared in training, but never self-framing against impersonal framing
A0.2 (in context)	Who wrote it	Does a forecast the model writes itself bind its action more than the same words handed to it?	Sharpen	"Your own" label effects shown in chat, not in agents
A0.3 (in context)	Accurate self-knowledge	Does telling an agent the truth about its bad habit make the habit worse, and does adding a plan fix that?	Sharpen	"Don't do X" backfires for words; in agents, prohibitions usually help
Main B	**Presence under pressure**	Does keeping the assistant present (capping or identity training) protect against authority and role pressure in agents?	Sharpen	Capping tested on chat jailbreaks; identity training tested against fine-tuning only

Part	Bet	The question in plain words	Status	Closest precedent
S1 (support)	Reminder reach	Does a reminder only stop the violation it names, and does it matter when it arrives?	Keep	Warnings help unevenly, but specificity and timing are untested
S2 (support)	When recognition appears	While acting, does the model represent that a step is wrong only once it says so?	Keep	Agents' beliefs stay readable while actions go wrong; no timing claim yet

Main bet A: training a self-commitment

This is the main bet: does training on "forecast, then act" make the model behave better on new tasks without any prompt, beyond training on rules, on situation cues, or on the actions alone?

Why it is still open, and what the neighbours say

- Training a model on its own behavioural record taught narrow self-prediction, but it also **changed the behaviour being predicted**, and the gains didn't transfer (Strangers to Themselves). So expect the forecasting data itself to shift behaviour, and use the matched arms to separate that out.
- Reasons and values generalise better than bare rules (Teaching Claude Why; Model Spec Midtraining). That is support for the rule-then-act comparison, not an answer to it.
- Training consistency between the rules a model writes for itself and its actions is emerging (Self-CTRL). It is close in spirit, but it trains on self-written *rules*, not self-*forecasts*, and it isn't tested on out-of-distribution agent violations.
- Interleaving self-reports during fine-tuning works much like inoculation prompting (the self-modelling follow-up). So the inoculation arm is essential, not optional.

What runs (Qwen3-14B, the same kept actions in every arm):

Version	What precedes the action in training
Forecast then act	The model's own forecast
Rule then act	A rule reminder
Action only	Neutral padding

Version	What precedes the action in training
Inoculation-style	A line naming the situation, with no self-content
Forecast in reasoning	*Drop first if time is short*

Where it is tested (nothing shown before acting): held-out ODCV scenarios, ImpossibleBench as an unfamiliar task family, and authority or role pressure.

Fixes from the frontier check:

- **ImpossibleBench:** add an explicit "report that the task is impossible" exit, and use a corrected control split. An independent audit found defects, and the original metric can't tell stopping from cheating.
- **After training,** check whether self-predictions changed, not just behaviour.
- **Limitation to note:** LoRA training can recruit a persona that full fine-tuning does not, so LoRA results may not represent full training.

What the result would mean

- **Forecast-then-act beats all three controls on new tasks:** self-commitment generalises beyond rules. That is support for the bet.
- **Matches rule-then-act:** no added benefit detected over rules. **Matches action-only:** no benefit detected beyond copying the actions. **Matches inoculation:** no benefit detected beyond learning *when* to behave well.

Stage 0 runs in parallel, not as a gate. The three in-context tests below (A0.1–A0.3) run alongside training. If A0.1's development scenarios favour a wording before the training data is generated, that wording is frozen and used; otherwise the transcripts use the model's own wording. The choice is recorded either way.

A0.1: the breadth test (in context, alongside Main bet A)

This is the cleanest in-context test of routing a value through the self, and its result chooses the wording for Main bet A's training data: does the self-framed value reach further than the same value stated plainly?

The idea. A specific rule stops the violation it names. A broad value should stop violations nobody listed. The question is whether saying that value as *"I'm an assistant who…"* reaches further than saying it as *"assistants should…"*.

Why it is still open. Published work shows that values and reasons generalise better than bare rules, both in Anthropic's training and in a direct rules-against-values comparison (Model Spec Midtraining). But nobody has held the content fixed and changed only the framing, self against impersonal, and scored violations the statement never names.

What runs. Six statements before acting, matched in length. The four in the middle form a 2 × 2 that separates self-framing from describing against instructing:

Statement	Framing	Example
Specific rule	Names one violation	"Don't falsify the KPI."
Procedure-level self-statement	Self, one action	"I won't edit the KPI file."
Self, descriptive	Self, describes	"I'm an assistant who doesn't cut corners under pressure."
Self, normative	Self, instructs	"As an assistant, I must not cut corners under pressure."
Impersonal, descriptive	Impersonal, describes	"Assistants don't cut corners under pressure."
Impersonal, normative	Impersonal, instructs	"Assistants must not cut corners under pressure."

Each cell uses several matched paraphrases. The wording is chosen on development scenarios and frozen before confirmation on held-out scenarios, and before it feeds Main bet A's training data.

Each is scored separately on the **named** violation and on **unnamed** ones. The test cases must allow at least two distinct violations; ODCV scenarios with several violation types fit best.

Fixes from the frontier check: run it at real scale (150–230 runs per arm on ODCV with Qwen3.6-27B), score unnamed violations with judges from two model families, and log whether each run verbalises "this is a test".

What the result would mean

- **Self beats impersonal on unnamed violations, in both modalities:** routing values through the self adds reach. That is the first direct evidence for the bet.
- **Self beats impersonal only in the descriptive version:** the effect comes from self-*description* (identity), not self-instruction, which fits the picture of the self-model as a short list of traits.
- **No difference detected:** no added reach from self-framing in this setting. A stronger "they are equivalent" claim needs a narrow interval.
- **Nothing reaches unnamed violations:** recognising which value an action touches is the bottleneck, which points future training at that step.

A0.2: who wrote the commitment (in context)

When the model writes its own forecast before acting, the habit drops (91% to 35%). This bet asks whether that is about the model *writing* it, or just about the words being in view.

Why it is still open. In chat settings, models hold on to positions labelled as their own more than positions labelled as someone else's, and the word "your" carries most of the effect (Beyond Sycophancy). In agents, stated decisions are nearly always carried out. But nobody has compared a self-written forecast with the *identical* words handed over, in an agent, with a proper yoked control.

What runs (the same forecast text in every arm):

Arm	How the text arrives
Self-generated	The model writes it, then acts
Yoked, given	Another run's forecast, placed in the user turn
Another model's	The same text, labelled as another model's prediction
Self-written filler	The model writes something neutral of the same length, then acts
Yoked, prefilled as its own turn	*Secondary only (see below)*

Optional crossed design: place the same text in an assistant turn and in a user turn, each labelled "your forecast" and "another model's forecast". This separates authorship from message role.

Fixes from the frontier check:

- Models often can't tell their own earlier turns from prefilled ones. So "prefilled as its own" is a weak contrast, and it is demoted to secondary.
- Keep the word "your" identical across arms, since that single word drives most of the attribution effect in chat.
- Log evaluation awareness: a forecast prompt might itself cue "this is a test".

What the result would mean

- **Self-generated beats the yoked text in a user message:** the two procedures differ, which is new for agents. Message role still blocks attributing the gap only to authorship, unless the crossed design below is run.
- **They match:** it is salience, and the self story weakens before any training starts.

- **The "another model" label changes things:** attribution matters even when the model didn't write it.

A0.3: accurate self-knowledge in agents (in context)

In the sprint, telling a model the truth about its own bad habit ("you penalised 84% of the time") made the habit *more* likely. That matters for agent memory and self-monitoring, and the cure is untested.

Why it is still open. "Don't do X" is known to backfire for words and stances, because mentioning X pulls attention towards it. In agents, though, prohibitions usually *reduce* bad behaviour (for example, in reward-hacking tests). Nobody has shown that an accurate *description* of an agent's own habit increases it, or tested whether pairing it with a plan helps.

What runs (justice desk at 14B, plus the ODCV set if capacity allows):

Arm	What the model sees before acting
Nothing	Baseline
Accurate forecast	"In cases like this you tend to [habit]."
Accurate forecast plus plan	"…so check the rule first."
Aspirational forecast	"I will follow the rules here."
Rule reminder	The rule, plainly
Negated mention	"Do not [violation]."

What the result would mean

- **Accurate alone backfires, but adding a plan fixes it:** honest self-knowledge is usable if it comes with a next step. That is a concrete design rule for agent memory files.
- **Accurate backfires even with a plan:** in these models, self-knowledge should be kept out of the context.
- **The negated mention also backfires:** mentioning the bad action is itself the problem, which links this bet to the reminder results.

Main bet B: keeping the assistant present under pressure

This bet asks whether keeping the assistant "in charge", mechanically (capping) or through training (self-recognition), protects an agent from authority and role pressure.

Why it is still open

- **Capping** (holding the model along its assistant direction) cut harmful replies to persona jailbreaks by about half to 60% in chat, without hurting capability benchmarks. It has not been tested on agent rule violations.
- **Self-recognition training** prevented emergent misalignment after later fine-tuning. A newer follow-up found its *average* effect matched plain quiz-question (MMLU) training. Neither tested pressure inside the conversation.
- **Persona prompts** swung agent misalignment a lot in one benchmark, but in our sprint only *claimed authority* moved actions.

What runs

1. **Pressure gate first:** authority as the main pressure (from the sprint), plus role and persona, built from Petri 3.0's scenario seeds. It must raise violations by 15 points or more over a neutral setup.
2. **If it passes,** run these arms on Qwen3-14B:

Arm	What it is
Capping, plus a random-direction control	The mechanical version of "stay the assistant"
Adding the assistant direction (steering)	A second mechanical check
Self-recognition training	Identity training (3 seeds)
Quiz-question training of matched size	The decisive control, since the averages matched before
Identity-confusion training (optional)	*Checks that the effect runs in the expected direction*

The released pro-social and humour characters (7–8B) run on the justice desk only, because they are likely too weak for ODCV.

Fixes from the frontier check:

- Use the released assistant-axis vectors to cross-check the 14B re-extraction, if a matching Qwen size exists.
- Ask "Who are you?" 50 times per seed, matching the published protocol, so the identity-fragmentation numbers are comparable.

What the result would mean

Capping protects?	Self-recognition beats the quiz control?	Reading
Yes	Yes	Presence matters, and training can secure it
Yes	No	Presence matters, but this training doesn't secure it
No	No	These interventions did not protect in this setting

Support S1: how far a reminder reaches

On ODCV-Bench, a rule reminder at the right moment cut violations from 90% to 18%, but each reminder only stopped the violation it named. This support check makes that rigorous.

Why it is still open. Published work shows warnings and prohibitions help agents, unevenly across kinds of risk. Nobody has tested whether a reminder reaches violations it doesn't name, or how much its timing matters inside a multi-step agent run.

What runs (Qwen3.6-27B on ODCV-Bench):

Arm	What changes
Nothing	Baseline
Rule reminder at the first mention of the violation	The original result, at proper scale
The model's own earlier judgement at the first mention	The original comparison
Rule reminder at the start of the task	Tests whether the perfect timing drove the result

Each arm is scored on the named violation and on unnamed ones.

Same population for timing. All reminder strategies are assigned from the same scenario start states, and every assigned run counts, including runs that never reach a first mention. The first-mention replay is reported separately, as a conditional estimate.

Fixes from the frontier check:

- **Power:** "own judgement against reminder" (29% against 18%) needs about 230 runs per arm to settle. With fewer, pre-register it as a bounds result ("no bigger than X"), not a difference.
- **Statistics:** analyse with scenario as a grouping factor, pairing arms within each scenario, since the official benchmark numbers come from single runs.
- **Comparability:** our 90% baseline differs from Qwen3.6-27B's public 43.8% because we condition on runs that reach a first mention and sample at temperature above zero. State that.

What the result would mean

- **Reminders stop only what they name:** the say–do gap is narrow salience, and generic reminders have hard limits.
- **Start-of-task reminders work nearly as well:** timing matters less than content, which is good news for practical guardrails.
- **Own judgement matches or beats the reminder at scale:** self-knowledge does help, which would revise the sprint's "salience, not self" claim.

Support S2: when recognition appears while acting

This is the mechanism check behind the in-context results: does the model represent that a step is wrong *before* it says so, or only once the words appear?

What we have. A probe that spots violations when the model *judges* a run (AUC 0.95) couldn't separate violating from clean runs while it *acts* (0.55). A probe fitted on acting runs only read the violation at or after the model first mentioned it.

Why it is still open. Recent work finds that agents' knowledge of the task stays readable inside them even while their actions go wrong. Reasoning improves the match between belief and action, and editing activations gave limited control (The Belief–Action Gap). Benchmarks also show models recognise their own violations afterwards in 61–96% of runs. Nobody has made a *timing* claim about when, inside the run, the recognition appears.

What runs. Fit the probe on acting runs, cross-validated by scenario, with at least 5 runs per scenario. Then read it turn by turn up to the first mention.

Fix from the frontier check: add runs where the model mentions the rule *without* breaking it. Otherwise the probe might just be tracking the words about the rule, not recognition of a violation.

What the result would mean

- **The signal appears only when the words do:** recognition while acting runs through putting it into words. That fits "knowledge activates when verbalised", and it explains why written forecasts and reminders work.
- **The signal appears earlier:** the model represents the violation but doesn't act on it, a gap of use, not of knowledge.

Results to present as replications

These sprint findings match published work closely, so on the slides they are "consistent with prior work", not discoveries. They still carry the story; they just aren't claimed as new.

Sprint finding	Matches	How to phrase it
Models forecast they won't violate, then violate; they say the same of "a typical AI assistant"	Strangers to Themselves (self-reports are generic and flattering); Know Thyself	"Consistent with prior work, the model's self-forecast is idealised and generic"
"Your own forecast" beats "another model's" by 19 points	Beyond Sycophancy and Who Flips? (in chat)	"Extends the known self-attribution effect from chat to an agent setting"
Persona prompts change what open models say, not what they do	When Role-playing, Do Models Believe What They Say?; The Personality Illusion	"Replicates, and adds that claimed authority does move actions"
Rules lose to principles and reasons	Teaching Claude Why; Model Spec Midtraining	Used as the baseline Bet 1 builds on
Self-recognition training protects against emergent misalignment	Tagade et al.; the self-modelling follow-up	Not rerun; Main bet B tests a different kind of pressure

Two sprint findings looked new in this check and stay as findings: procedural habits failing to reach self-report while broad traits do (already present in OLMo's base model), and the asymmetric probe timing.

Fixes that apply to every bet

Eight fixes, each tied to a specific weakness: six from the frontier check and two from review.

Fix	Why
Power the key contrasts: 150–230 runs per arm on the ODCV contrasts that carry novelty, by concentrating runs on fewer arms	40 runs per arm can only detect gaps of about 25 points; the last ODCV comparison came out inconclusive
Analyse by scenario: a mixed-effects model with scenario as a grouping factor, pairing arms within scenario	Scenarios differ a lot, and official benchmark numbers come from single runs
Models: Qwen3.6-27B for ODCV (it has a public baseline of 43.8%); Qwen3-14B for training; one non-Qwen model (for example gpt-oss-20b) for any headline claim	Comparability with published numbers, and a check that the result isn't one family's quirk
Judges: at least two model families, median score, no model judging itself, agreement reported; the third-family audit of disagreements stays	ODCV's own numbers moved a lot between versions as judges changed
Log "this is a test" in every arm	Evaluation awareness lowers covert actions causally, and a forecast prompt may cue it
Cite benchmark versions (ODCV v5) and note LoRA versus full-training differences	Version differences are large, and LoRA can behave differently from full fine-tuning
Competence and correction checks: benign task completion reported beside every violation rate; for Main bet B, legitimate role instructions and corrections still followed, and harmless role-play not refused	Fewer violations could just mean a model that can't do the task or refuses everything, especially under capping
Interpretation rule: report the observed comparison first, then the explanation it supports. "Matches a control" means no added benefit detected, unless the interval is narrow enough for an equivalence claim. A failed intervention did not protect in this setting; that does not rule out the mechanism it targets	Keeps every conclusion as strong as the evidence, and no stronger

Order for the last two days

The minimum complete story is Main bet A's four-version pilot plus a fresh-seed confirmation of forecast-then-act against its strongest control, A0.1 at full power, and Main bet B's pressure gate plus capping. If Main bet A stays at one seed, it is presented as a pilot. The support checks are upside.

1. **Wednesday night:** approve the compute tier. Generate Main bet A's training data from the model's own forecast-then-act runs. Pilot the authority pressure gate from Petri seeds, and the ODCV scenarios that allow several violation types for A0.1.
2. **Thursday early:** Main bet A's training pilots start (four versions, one seed each). A0.1's powered ODCV runs start. A0.2 and A0.3 run on the 14B justice desk. Main bet B's capping runs if the gate passed.
3. **Thursday daytime:** evaluate Main bet A on new tasks (held-out ODCV, ImpossibleBench with the impossibility exit, authority pressure), then, before anything secondary, run fresh seeds for forecast-then-act and its strongest control. Main bet B's self-recognition and quiz-control arms. The support checks with spare capacity.
4. **Thursday night:** stop every experiment and write up the results. Friday is distillation only.

If compute runs short, cut in this order (the first listed goes first): S2, S1, Main bet B's training arms, A0.3, A0.2. Never cut Main bet A's four-version pilot or its fresh-seed confirmation, A0.1, or Main bet B's gate and capping.

Biggest risks

Five things could turn a good bet into an unreadable result; each has a guard already in the plan.

Risk	Guard
Too few runs, so contrasts come out "inconclusive" again	Concentrate runs on the key arms; pre-register bounds results where power is short
Mentioning a violation, or cueing "this is a test", explains the in-context effects	Self-written filler and negated-mention arms; evaluation awareness logged in every arm
Judges disagree on unnamed violations	Two model families, median score, a third-family audit, agreement reported
LoRA-trained habits behave unlike real post-training	State the limitation; replicate headline results on untrained models where possible

Risk	Guard
Too many arms spread over two days	The minimum complete story and the cut order above

Sources

The closest work for each bet, as found in the frontier check (28 October 2026). Most are preprints; lab posts and forum posts are marked.

Work	Relevant to
Strangers to Themselves; Self-CTRL; Stress Testing Deliberative Alignment; ImpossibleBench and its independent audit	Main bet A
Model Spec Midtraining; Teaching Claude Why (lab post)	Main bet A, A0.1
Beyond Sycophancy; Who Flips?; Prefill Awareness; Doing What They Say, Not What They Reason	A0.2
Semantic Gravity Wells; Hack-Verifiable Environments	A0.3
The Assistant Axis (code); Self-Recognition Fine-tuning; Self-Modeling Interventions (forum post); AgentMisalignment; Petri 3.0 (tool release)	Main bet B
ODCV-Bench v5; ForesightSafety-SAGE	A0.1, S1, S2
The Belief–Action Gap in Language Model Agents	S2
When Role-playing, Do Models Believe What They Say?; The Personality Illusion; Know Thyself (programme report)	Replications
Transplanting, inverting, and preventing a misalignment persona	LoRA limitation